

Pharmaceutical Report

Group 22

Executive Summary

Table of Contents

Background and Data

What is dementia and the importance of pharmaceutical treatment

Importance of equity across socioeconomic status

Pharmaceutical collection background

Note that dispensing doesn't immediately imply that the medication was taken

Association of dispensing between deprivation and time

Background on the NZDep2023 population counts by deprivation decile and health district, and the attributes included + suppression rules and unknown districts

The pharmaceutical data is a repeated cross sectional data frame showing how medicine was distributed in different regions over different years.

Note that just because medicine was dispensed does not mean that it was taken.

Additional district level variables to look into:

Income, age, employment, education, racial profile (Probably not), family dependencies.

Ethics, Privacy and Security

- Ethical considerations:

- Any comparison of groups being done in this project shouldn't cause have any implications about financial status between said groups. For example, Māori people take less medication than Pākehā because they can't afford them.
- Given the nature of the data collection, no causal conclusions can be drawn. The analysis will only be able to create commentary about the trends and relationships between the variables.

- Privacy considerations:

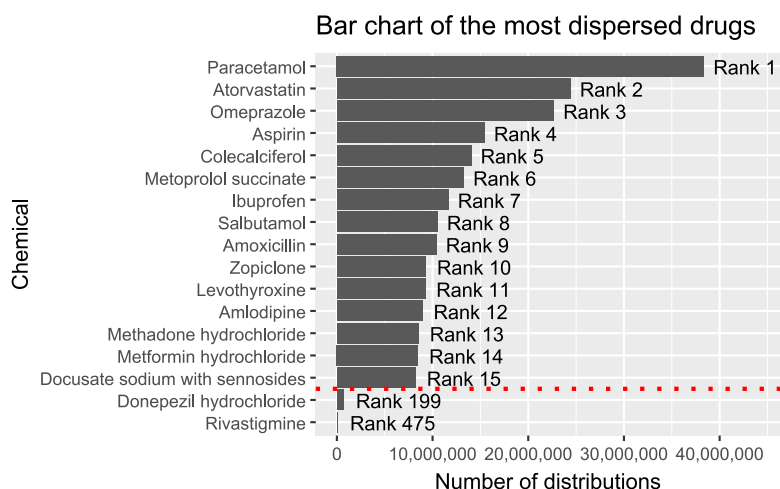
- We should not be removing the suppression layer from the count of medication distribution. While there is an oversight in how this data is provided that allows users to do this, and there

is code demonstrating that this is possible in this project, the values were randomly adjusted so that for the example instance doesn't have the true count revealed.

- For all analysis, the suppressed values will be imputed. Currently the method of replacing these with the median of the suppression range.

Exploratory Data Analysis

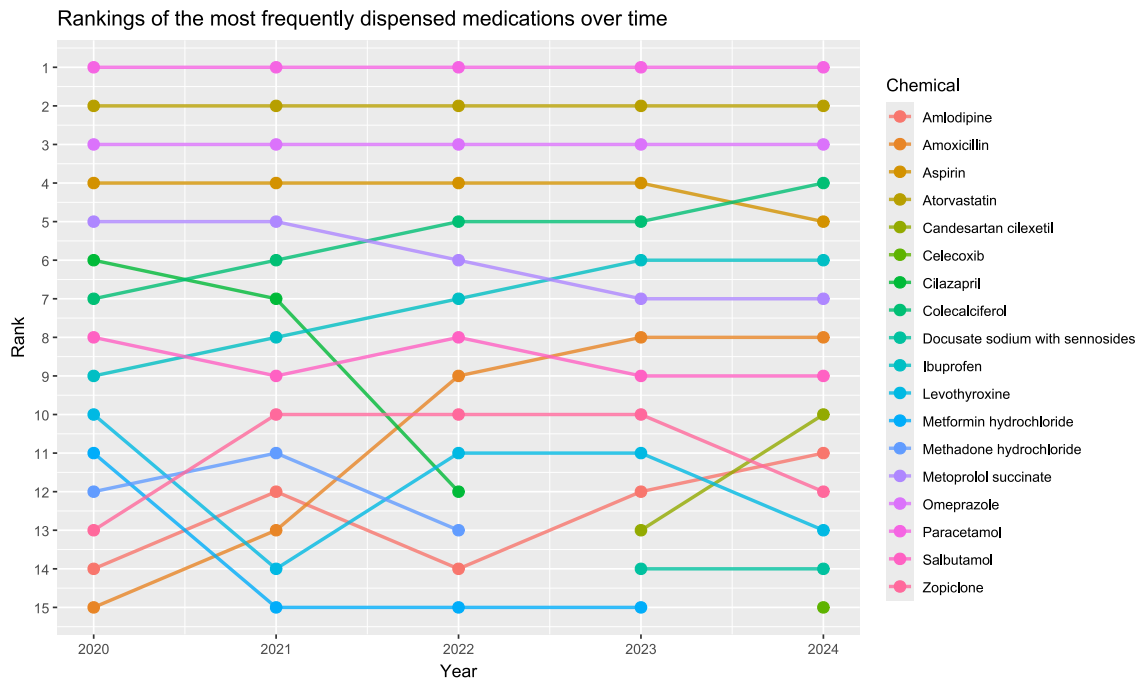
The graph below shows an unequal distribution of dispensing of medications of the 15 most widely distributed chemicals. Paracetamol is by far the most dispensed, with over double the amount of distributions compared to the aspirin, the chemical in 4th place. The second and third most widely distributed medications are Atorvastatin and Omeprazole. Overall, the distribution has a long tail. Donepezil hydrochloride and Rivastigmine aren't highly distributed enough to make the top 15, with the former ranking 199th across the 5 years of data, and the latter ranking 475th over the same period.



Looking at the line chart below showing rank over time, it can be seen that there are no changes to the rank of the top 3 medicines across the five years. Moving down the rankings, more variability is displayed. Colecalciferol rose from 7th to 4th, metoprol succinate moved down from 5th to 7th and Ibuprofen from 9th to 6th. This relates to two things:

- The long tailed nature of the ranked distribution means that the steeper decreases in medication top of the ranks are rank unphased relative to any changes over time.
- The similar distributions over time between subsequently ranked medications are more likely to have their ranks changed by upward or downward trends in distribution over time.

This suggests that any future modelling would benefit from having year of distribution as a regressor, but it would be less likely to change the rankings of the highest distributed drugs



The table below shows the distribution count and ranks of the two dementia medications of interest. There has been an upwards trend in distribution for both of these medications, while rivastigmine has had its rank reduced in recent years. This is because while the distribution of this medication has increased in these years, not as much as other medication that had a lower distribution count that rivastigmine in 2021.

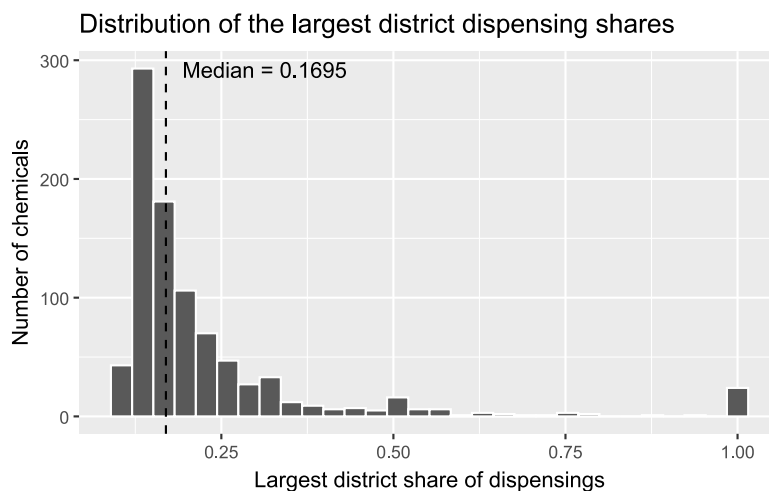
``summarise()`` has grouped output by 'YearDisp'. You can override using the ``.groups`` argument.

Year	Donepezil hydrochloride distribution count	Rivastigmine distribution count
2020	119502	6552
2021	121298	7545
2022	126223	7270
2023	137048	7688
2024	151711	8399

Donepezil hydrochloride distribution rank	Rivastigmine distribution rank
203	473

Donepezil hydrochloride distribution rank	Rivastigmine distribution rank
202	459
205	465
198	472
191	476

It is useful to know how frequently medications are distributed in a high concentration to a few health districts relative to an even spread across all health districts. One way to measure this is by using the proportion of distributions going toward the health district where most of the distributions occur. There are 19 health districts, and if the distributions occurred equally across said districts, this metric would take the value of 1/19. In the opposite extreme, if the medication was distributed in only 1 district, the metric would take the value of 1. Below is a histogram showing how each of the medications fall on this proportion scale.

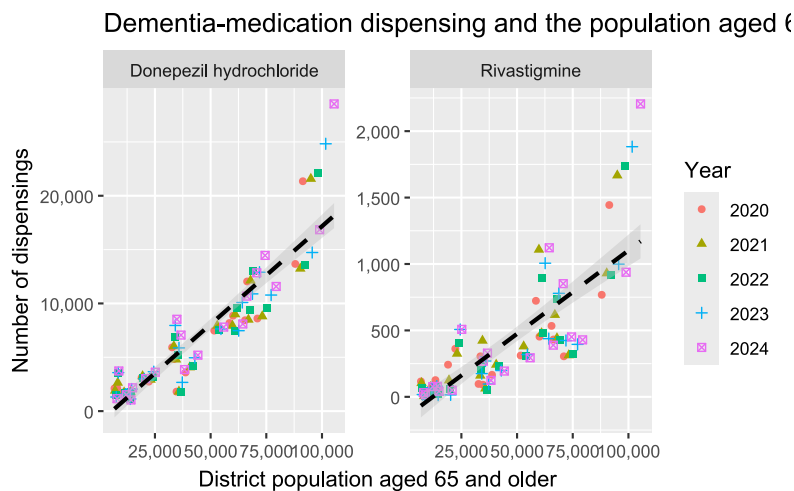


The median value is 0.1695, meaning that a typical medication will have roughly 17% of their medication go towards the district that leads in distribution. The lowest observed proportion is 0.1049839 meaning that no medication is equally distributed across all districts. There's a small group of medications that have a proportion value of 1, meaning that they are only distributed in a single district. Note that this metric doesn't speak to the number of distributions for these medications, and are likely very niche with an estimated mean of 12.38 and max of 70. The medications with these lower distributions are either very niche, available from retailers outside of pharmacies or not recorded in some districts.

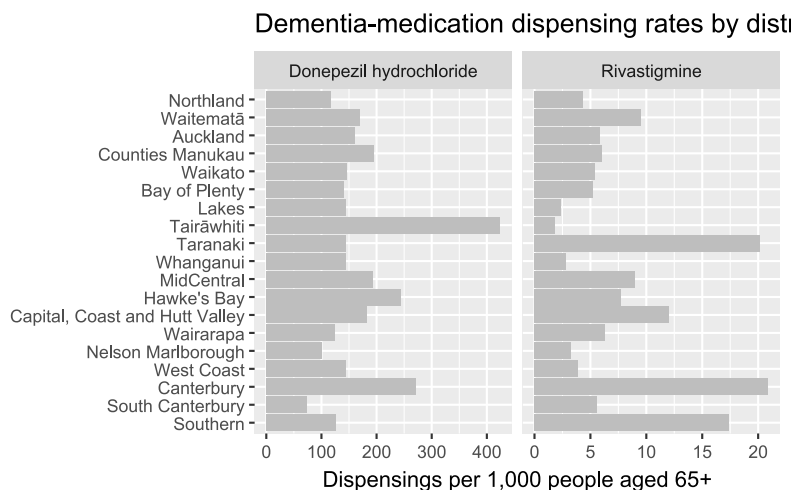
Dementia as a risk is strongly associated with old age. This makes the districts with a higher count of aged population more relevant to this analysis than total population size in the health district. The graphs below show a positive relationship between aged population count and distributions of the two key dementia medications. This relationship is stronger for Donepezil Hydrochloride,

but both show that there are some observations above the line of best fit on the higher end of the district aged population. These observations belong to the `{r}`dementia_population` |> group_by(Chemical) |> slice_max(order_by = NumDisps, n = 5) |> pull(District) |> unique()` district, and as the years increase it can be seen diverging further away from the line of best fit. Overall, this high correlation means that the older population size accounts for a large amount of the inter-district variation.

```
`geom_smooth()` using formula = 'y ~ x'
```

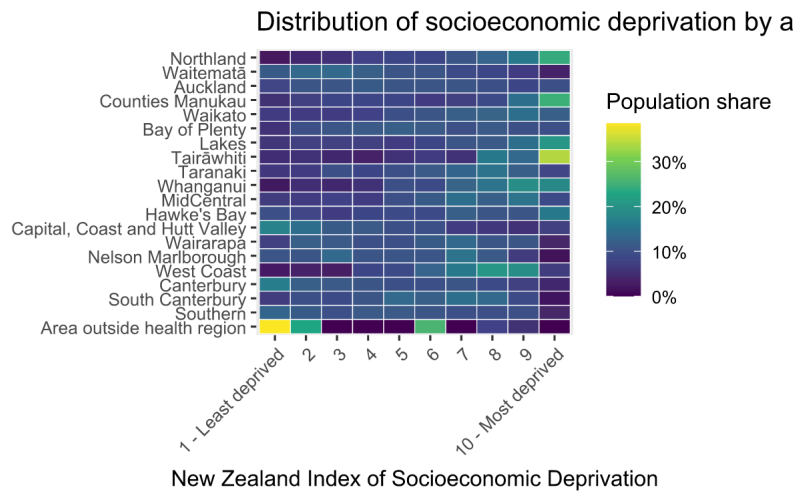


Crude rates were calculated as annual distributions per 1,000 people aged 65 years and over for the 2024 year. The table below shows the variation in these distribution rates across the different districts.



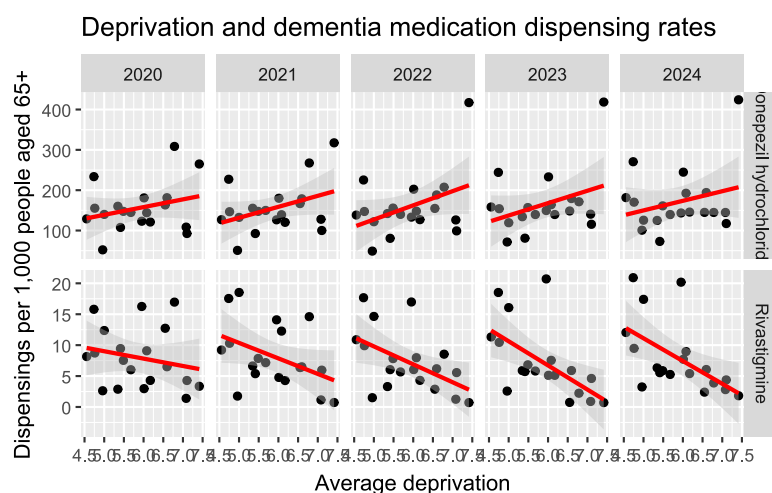
The first observation of note is that donepezil hydrochloride has a much higher rate than rivastigmine. Tairāwhiti has the highest overall rate when you take into account said larger rates

for donepezil hydrochloride, as it is the district that has the highest rate for this medication. Canterbury and Hawke's Bay have the second and third highest rates respectively. Alternatively, Rivastigmine has the highest concentration of distribution in Canterbury, Taranaki and Southern. Taranaki doesn't stand out for being high or low with donepezil hydrochloride, and Tairāwhiti has the lowest distribution rate for rivastigmine. This could be related to the high proportion of the population of Tairāwhiti falling higher on the latest (2023) NZ deprivation scale as seen below.



The plots below show how the average distribution rate per aged person changes for the two dementia medications over the deprivation index. For donepezil hydrochloride, the trend in the relationship between the distribution rate and the deprivation index increases as time increases, peaking at 2022, followed by a decreasing trend in subsequent years. Alternatively, these trends have consistently steeper downward trends as the years increase. This is a strong argument to test interaction effects between year and deprivation index when predicting the number of medication distributions.

```
`geom_smooth()` using formula = 'y ~ x'
```



These results remain exploratory. There are only about 19 districts per year, the same districts are observed repeatedly, and district deprivation changes little over the study period. Confidence intervals are wide, especially for rivastigmine, and the slopes may be sensitive to influential districts such as Tairāwhiti, Canterbury or Taranaki.

Detailed Analysis Results

A variety of parametric models are tested to find which most appropriately fit the data. Creating the appropriate model allows for inference on the relationships between the different regressors. The table compares models that predict the number of dementia medication distributions. The Poisson models assume that the variance is equal to the mean, while the negative binomial models allow for additional variation. The mixed models also include a random intercept for district, which accounts for repeated observations and differences between districts that are not explained by the other variables. The models become more complex as interactions are added, allowing the relationships with year and deprivation to differ between medications. The final model also uses the population aged 65 years and over as an offset, allowing it to model dispensing rates rather than only raw counts.

```
# A tibble: 1 × 4
  residual_deviance residual_df deviance_ratio pearson_dispersion
      <dbl>         <int>         <dbl>         <dbl>
1    426944.         188         2271.         2574.
```

```
# A tibble: 1 × 4
  residual_deviance residual_df deviance_ratio pearson_dispersion
      <dbl>         <int>         <dbl>         <dbl>
1    313303.         186         1684.         1684.
```

```
# A tibble: 1 × 4
  residual_deviance residual_df deviance_ratio pearson_dispersion
      <dbl>         <int>         <dbl>         <dbl>
1      313260.         185         1693.         1693.
```

```
# A tibble: 1 × 4
  residual_deviance residual_df deviance_ratio pearson_dispersion
      <dbl>         <int>         <dbl>         <dbl>
1      313253.         184         1702.         1702.
```

```
# A tibble: 1 × 4
  residual_deviance residual_df deviance_ratio pearson_dispersion
      <dbl>         <int>         <dbl>         <dbl>
1      309410.         183         1691.         1684.
```

model	explanation	df	BIC
mixed_nb_4	Adds the population aged 65 years and over as an offset, producing population-adjusted dispensing rates.	9	2897
mixed_nb_3	Also allows the deprivation relationship to differ between medications.	9	2926
mixed_nb_1	Negative binomial mixed model with a year-by-deprivation interaction.	7	2973
mixed_nb_2	Also allows the time trend to differ between medications.	8	2977
nb_1	Negative binomial version of poisson_1, allowing for overdispersion.	5	3147
nb_4	Negative binomial version of poisson_4, allowing for overdispersion.	8	3151
nb_2	Negative binomial version of poisson_2, allowing for overdispersion.	6	3152
nb_3	Negative binomial version of poisson_3, allowing for overdispersion.	7	3157
mixed_poisson	Poisson model that also accounts for differences between districts.	6	15635
poisson_4	Also allows the deprivation relationship to differ between medications.	7	311105

model	explanation	df	BIC
poisson_3	Also allows the time trend to differ between medications.	6	314943
poisson_2	Allows the relationship between deprivation and dispensing to change over time.	5	314944
poisson_1	Adds year and average deprivation as separate effects.	4	314982
poisson_0	Poisson model containing differences between the two medications only.	2	428612

The best fitting model, based on having the lowest BIC, uses the number of distributions as the response. It treats the number of distributions as a negative binomial distributed variable that depends on several district-level regressors. These are year, average deprivation and medication. Interactions between these variables are also included. The log of the population aged 65 years or older is used as an offset, which accounts for differences in the size of the older population across districts. District is included as a random intercept because each district is observed multiple times.

The model suggests that the distribution rates for the two medications changed differently over time. At the average level of deprivation, the coefficient for year is 0.0293 for donepezil hydrochloride. After exponentiating this coefficient, the estimated rate ratio is approximately 1.030. This suggests that the distribution rate for donepezil hydrochloride increased by around 3.0% per year. The year coefficient for rivastigmine also includes the interaction between year and chemical. This gives a coefficient of $0.0293 - 0.0629 = -0.0336$. The corresponding rate ratio is approximately 0.967, suggesting that the distribution rate for rivastigmine decreased by around 3.3% per year. This means that the two medications had different patterns over time.

The relationship between deprivation and distribution also differs between the medications. The deprivation coefficient cannot be interpreted directly because year has not been centred. The deprivation effect for a particular year can be found by combining the deprivation coefficient with the interaction between year and deprivation.

For donepezil hydrochloride in 2020, the estimated deprivation coefficient is approximately 0.245. This gives a rate ratio of approximately 1.28. Therefore, a one-unit increase in average district deprivation was associated with an estimated 28% increase in the donepezil hydrochloride distribution rate in 2020, after accounting for the other variables in the model. For rivastigmine, the interaction between deprivation and chemical also needs to be included. This gives an estimated deprivation coefficient of approximately -0.308 in 2020 and a rate ratio of approximately 0.735. Therefore, a one-unit increase in average district deprivation was associated with an estimated 26.5% decrease in the rivastigmine distribution rate. This supports the pattern seen in the exploratory analysis, where the deprivation relationships moved in opposite directions for the two medications.

The interaction between year and deprivation is negative. This suggests that the relationship between deprivation and donepezil hydrochloride distribution became less positive over time. It also means that the negative relationship between deprivation and rivastigmine distribution became stronger over time.

The random intercept for district has a standard deviation of 0.357. Exponentiating this gives approximately 1.43. This means that a district one standard deviation above the average district effect had an expected distribution rate around 43% higher, after accounting for the variables included in the model. There is therefore still variation between districts that is not explained by year, deprivation, medication or older population size.

The negative binomial dispersion parameter is 7.33. This allows the variance of the response to be larger than its mean. This is more suitable than a Poisson model because the earlier Poisson models showed substantial overdispersion (as was reflected in the poor BIC fit).

The large values of some coefficients, such as the intercept and deprivation coefficient, are mainly caused by using the calendar year directly. The intercept currently represents the expected value in year zero.

```
Formula:
NumDisps ~ Year * deprivation_c + Year * Chemical + deprivation_c *
  Chemical + offset(log_population_65_plus) + (1 | District)
```

```
Data: dementia_model_data
      AIC      BIC    logLik -2*log(L)  df.resid
  2868.250  2897.473 -1425.125  2850.250      181
Random-effects (co)variances:
```

```
Conditional model:
Groups   Name      Std.Dev.
District (Intercept) 0.3571
```

```
Number of obs: 190 / Conditional model: District, 19
```

```
Dispersion parameter for nbinom2 family (): 7.33
```

```
Fixed Effects:
```

```
Conditional model:
              (Intercept)              Year
              -61.01661              0.02926
      deprivation_c              ChemicalRivastigmine
              79.22679              123.95147
      Year:deprivation_c              Year:ChemicalRivastigmine
              -0.03910              -0.06287
```

```
deprivation_c:ChemicalRivastigmine  
-0.55302
```

A limitation of this model is that while most of the attributes are available on an annual level, notably deprivation index is not. An improvement would be to apply the deprivation index on the year that's appropriate, but the data informing this deprivation metric isn't sourced annually. Instead, it is sourced from the NZ census which is run every 5 years.

Another limitation is that the data for distribution of medication is shared at the district level as opposed to the individual level. While a more granular view of the attributes would improve the accuracy of the model, and create stronger inferences, it becomes a violation of privacy if individuals medication status were available publicly on the individual level. Also to do with data availability, the model creates a relationship with the number of medication distributions with the regressors, as opposed to the medication taken by individuals which may not be as relevant to interested parties.

Conclusions and Recommendations